

Aurora: A Hybrid Quantum Algorithm for Fuel-Optimal Aircraft Tail Assignment via Superposition, Entanglement, and Interference

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Abstract

Fuel is the largest controllable cost in commercial aviation, and the decision of which physical aircraft flies each leg of a network—the *tail assignment problem*—is an NP-hard combinatorial optimization whose feasible space grows exponentially in the number of candidate aircraft rotations. We present **Aurora**, a hybrid quantum–classical algorithm that couples three quantum kernels around a classical decomposition and variational loop. Aurora casts tail assignment as an exact-cover QUBO, maps it to an Ising cost Hamiltonian, and solves it with the Quantum Approximate Optimization Algorithm (QAOA) using a constraint-preserving XY mixer. Two auxiliary kernels carry *provable* quadratic speedups: quantum amplitude estimation prices each rotation by its wind-dependent *expected* fuel burn in $\mathcal{O}(1/\varepsilon)$ rather than the Monte Carlo $\mathcal{O}(1/\varepsilon^2)$, and amplitude amplification prepares exact-cover-feasible superpositions in $\mathcal{O}(\sqrt{N/M})$ rather than $\mathcal{O}(N/M)$. We give the circuit-level construction, make explicit where superposition, entanglement, and interference each act, and back the design with *noiseless statevector simulations* on synthetic exact-cover instances: layerwise-optimized QAOA improves the mean approximation ratio from 0.88 at $p=1$ to 0.95 at $p=8$ while the optimum-sampling probability rises roughly nine-fold; instance hardness tracks problem-graph connectivity; and the amplitude-estimation and feasibility kernels reproduce their expected quadratic scaling. We provide an honest ledger separating the proven quadratic advantages of the auxiliary kernels from the empirical, instance-dependent advantage of the QAOA optimizer, and an illustrative case study on a synthetic mid-size carrier. All quantitative results are simulation-derived or modeled projections, not measurements on a production fleet or on quantum hardware.

1 Introduction

Commercial aviation spends a very large fraction of its operating cost on jet fuel; the International Air Transport Association (IATA) has reported year-on-year swings in the global industry fuel bill on the order of one hundred billion U.S. dollars, with fuel routinely representing roughly a quarter to a third of a carrier’s operating expense [8]. Figure 1 places recent figures in context. A low-single-digit percentage im-

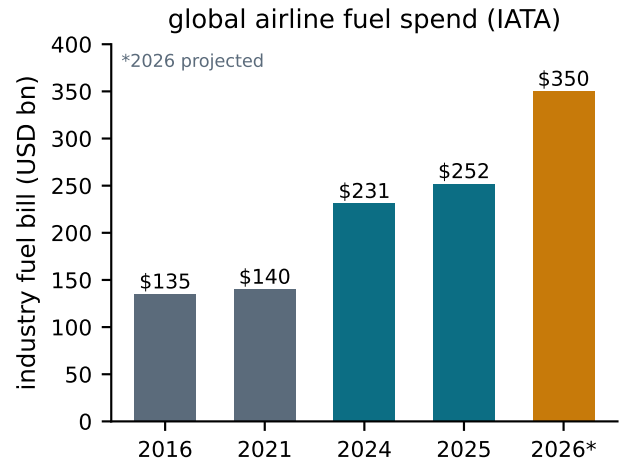


Figure 1: Global airline industry fuel bill, IATA-reported anchor points; 2026 projected [8]. Fuel is the largest controllable line item in airline operating cost, so small percentage efficiencies are large absolute sums.

provement in fuel efficiency therefore translates into a saving measured in hundreds of millions of dollars and tens of thousands of tonnes of CO₂ per operator per year.

A substantial share of that efficiency is decided at the planning desk, in the *tail assignment problem*: given a set of flight legs that must all be flown and a fleet of individual aircraft (*tails*), decide which tail flies each leg, and along which vertical/lateral profile, so as to minimize total fuel subject to airport continuity, minimum turnaround, curfew, and maintenance constraints [1, 7]. The decisions are strongly coupled: committing a tail to one leg removes it from every other leg it could have flown that day. The problem is NP-hard and is solved in practice by mixed-integer programming (MIP) with branch-and-price, which is excellent but, on a full network, typically exhausts its time budget before closing the optimality gap and returns the best incumbent found.

Quantum optimization offers a different handle on the exponential space. The Quantum Approximate Optimization Algorithm (QAOA) [3, 19] represents all 2^n candidate configurations of n binary variables as a single n -qubit super-

position and shapes their amplitudes so that low-cost configurations become likely to be measured. Vikstål *et al.* [1] demonstrated that tail assignment reduced to *exact cover* can be solved by QAOA with near-unit success probability on real airline instances of up to 25 qubits, and subsequent work has extended the route-optimization framing [2]. Our contribution is not to reinvent that core but to package it into a deployable pipeline, to be precise about *where* quantum advantage is and is not established, and to characterize the design empirically in simulation.

Contributions.

1. We formulate fuel-optimal tail assignment as an exact-cover QUBO with a wind-aware, risk-adjusted rotation cost, and derive its Ising cost Hamiltonian, making explicit that the entangling $Z_r Z_{r'}$ couplings are exactly the pairs of rotations that compete for a shared leg (§4).
2. We specify **Aurora**, a hybrid pipeline of three quantum kernels—amplitude-estimation cost, amplitude-amplified feasibility, and XY-mixer QAOA—wrapped in classical decomposition and parameter optimization (§5).
3. We give an explicit map of the “quantum trinity”—superposition, entanglement, interference—onto concrete circuit elements (§3).
4. We evaluate the design with noiseless statevector simulations (§6–§7): depth scaling, instance hardness vs. connectivity, amplitude-estimation and feasibility speedups, penalty calibration, resource growth, and the $p=1$ energy landscape.
5. We provide an honest complexity ledger (§9), an illustrative case study (§10), and a candid threats-to-validity discussion (§12).

2 Background and Related Work

NISQ optimization. Present-day devices are noisy and of intermediate scale [9, 17]. Variational quantum algorithms [10] sidestep deep coherent evolution by delegating parameter search to a classical optimizer, and QAOA [3] is the canonical variational algorithm for combinatorial optimization. Its performance, mechanism, and parameter-transfer behavior were characterized by Zhou *et al.* [11], who introduced the layerwise INTERP initialization we adopt in §6; Google demonstrated QAOA on non-planar graphs on a superconducting processor [12]. Whether QAOA yields an asymptotic speedup for NP-hard optimization remains open; Guerreschi and Matsuura [18] argue that hundreds of qubits are needed before QAOA competes with classical solvers on MaxCut, a caution we take seriously in §9.

Constraint handling. Penalty terms convert constrained problems to QUBOs but enlarge the search space with infeasible states. Hadfield *et al.* [4] generalized QAOA to a Quan-

tum Alternating Operator Ansatz with constraint-preserving mixers (e.g., the XY mixer) that confine evolution to a feasible subspace; warm-starting [14] and CVaR objectives [13] further improve variational optimization. Aurora uses an XY mixer within each leg’s competing-rotation register (§5.3).

Amplitude amplification and estimation. Grover search [20] and its generalization, amplitude amplification and estimation [5], deliver provable quadratic speedups for unstructured search and for Monte Carlo expectation estimation [6]. These primitives underpin quantum risk analysis [15] and admit phase-estimation-free variants [16] better suited to shallow devices. Aurora uses estimation to price stochastic fuel burn and amplification to prepare feasible states (§5.1–§5.2).

Airline optimization. Tail assignment and fleet routing are classical operations-research problems [7]. The quantum treatment closest to ours is Vikstål *et al.* [1], who reduce tail assignment to exact cover and solve it with QAOA; [2] extends the route framing. Aurora differs by adding risk-aware cost estimation and feasibility amplification around the QAOA core and by an explicit accounting of advantage.

3 The Quantum Trinity as a Design Principle

Aurora uses three physical resources, each for the task it is suited to. We state the mapping here because every circuit below is an arrangement of these three.

- **Superposition.** A layer of Hadamard gates places the register in an equal blend of all 2^n candidate schedules. The search space is *loaded*, not enumerated.
- **Entanglement.** Two-qubit $Z_r Z_{r'}$ rotations couple rotations that compete for the same leg, so the register can represent constraints *between* decisions, not just the decisions themselves.
- **Interference.** Alternating cost-phase and mixing operators make amplitude paths to low-fuel schedules add in phase and paths to high-fuel schedules cancel. This is the computation.

In one sentence: superposition makes every candidate present at once, entanglement writes the fuel-and-constraint structure into the relative phases, and interference lets the physics cancel the bad schedules and amplify the good one.

4 Problem Formulation

4.1 Exact-cover model

Following [1, 7], we precompute classically a large set R of feasible *rotations*: legal sequences of legs a single aircraft can operate in a planning period, each already respecting turnaround, continuity, curfew, and maintenance rules. We assign one qubit to each rotation $r \in R$, with binary variable

$x_r \in \{0, 1\}$ selecting it. Each rotation carries a fuel cost c_r , and each leg f in the leg set F must be covered by exactly one selected rotation:

$$\begin{aligned} \min_{x \in \{0,1\}^{|R|}} \quad & \sum_{r \in R} c_r x_r \\ \text{s.t.} \quad & \sum_{r \ni f} x_r = 1 \quad \forall f \in F, \end{aligned} \quad (1)$$

where $r \ni f$ denotes “rotation r covers leg f .”

4.2 QUBO and Ising mapping

We fold the exact-cover constraint into the objective with a penalty weight $\lambda > 0$ chosen large enough that any coverage violation outweighs any fuel saving:

$$C(x) = \sum_{r \in R} c_r x_r + \lambda \sum_{f \in F} \left(1 - \sum_{r \ni f} x_r\right)^2. \quad (2)$$

Substituting $x_r = \frac{1}{2}(1 - Z_r)$, with Z_r the Pauli-Z operator on qubit r (eigenvalues ± 1), turns (2) into a diagonal Ising cost Hamiltonian

$$H_C = \sum_{r \in R} h_r Z_r + \sum_{r, r' \in S_f} J_{rr'} Z_r Z_{r'} + \text{const.} \quad (3)$$

Expanding the squared penalty in (2) for a single leg f with competing-rotation set $S_f = \{r : r \ni f\}$ and $k_f = |S_f|$, using $Z_r^2 = \mathbb{I}$, gives

$$\begin{aligned} \left(1 - \sum_{r \in S_f} x_r\right)^2 &= \frac{1}{4} \left[\sum_{r, r' \in S_f} Z_r Z_{r'} \right. \\ &\quad \left. + 2(2 - k_f) \sum_{r \in S_f} Z_r + (2 - k_f)^2 \right], \end{aligned} \quad (4)$$

so that every pair of rotations sharing at least one leg acquires a coupling

$$J_{rr'} = \frac{\lambda}{2} |\{f : r \ni f \text{ and } r' \ni f\}| \quad (5)$$

and each rotation acquires a local field $h_r = -\frac{1}{2}c_r + \lambda \sum_{f \ni r} \frac{1}{2}(2 - k_f)$.

Remark 1. Equation (5) is the crux of the construction: the $Z_r Z_{r'}$ terms—and hence all entanglement in the circuit—appear exactly for pairs of rotations that compete for a shared leg. The connectivity of this problem graph both determines the entangling gate pattern and, empirically, predicts instance hardness (§7.2), consistent with [1].

Worked micro-example. Consider three rotations $\{r_1, r_2, r_3\}$ and two legs $\{f_1, f_2\}$ with $r_1 \ni \{f_1\}$, $r_2 \ni \{f_1, f_2\}$, $r_3 \ni \{f_2\}$ and costs $c = (1.0, 1.8, 1.1)$. Leg f_1 is contested by $\{r_1, r_2\}$ and f_2 by $\{r_2, r_3\}$, so by Eq. (5) the Ising model has exactly two entangling edges, $J_{12} = J_{23} = \lambda/2$, and no J_{13} term— r_1 and r_3 never compete. The unique exact cover is $\{r_1, r_3\}$ (cost 2.1), which the penalized QUBO (2) selects as its global minimum for $\lambda \gtrsim 1.5\bar{c}$ (§7.5). This three-qubit instance is the pattern the circuit of Fig. 2 entangles and the depth study of §7.1 scales up.

4.3 Risk-aware rotation cost

A rotation’s fuel burn is not deterministic: it depends on winds aloft, payload, and temperature. We therefore define c_r as an expectation over an uncertainty model $\mathcal{W}$ (historical plus forecast wind fields),

$$c_r = \mathbb{E}_{w \sim \mathcal{W}} [\text{fuel}_r(w)], \quad (6)$$

which we estimate with the amplitude-estimation kernel of §5.1.

5 The Aurora Algorithm

Aurora is a hybrid loop: classical decomposition and parameter search on the outside, three quantum kernels on the inside.

5.1 Kernel 1 — Risk-aware cost by amplitude estimation

To evaluate (6), we encode $\mathcal{W}$ via a state-preparation unitary $\mathcal{A}$ and encode $\text{fuel}_r(w)$ into the amplitude of an ancilla, so that the ancilla’s probability of being $|1\rangle$ equals the (normalized) expected fuel. Quantum amplitude estimation [5, 6, 16] returns this expectation to additive error ε using $\mathcal{O}(1/\varepsilon)$ applications of $\mathcal{A}$, a quadratic improvement over the $\mathcal{O}(1/\varepsilon^2)$ samples required by classical Monte Carlo. This speedup is proven and unconditional; §7.3 reproduces it numerically.

5.2 Kernel 2 — Amplitude-amplified feasibility

A layer of Hadamards prepares $|s\rangle = |+\rangle^{\otimes n}$ over all 2^n selections. A feasibility oracle O_f marks the exact-cover-valid states. Amplitude amplification [20, 5] rotates amplitude into the feasible subspace: if M of the $N = 2^n$ states are feasible, roughly $\frac{\pi}{4}\sqrt{N/M}$ Grover iterations suffice, versus the $\mathcal{O}(N/M)$ expected trials of classical rejection sampling—a proven quadratic speedup (§7.4). Kernel 3 then begins from a state concentrated on feasible schedules.

5.3 Kernel 3 — Optimization by interference

The core is a depth- p QAOA [3]. Starting from the (amplified) superposition, we apply p alternating layers of a cost operator and a mixer,

$$|\gamma, \beta\rangle = \prod_{j=1}^p e^{-i\beta_j H_B} e^{-i\gamma_j H_C} |s\rangle, \quad H_B = \sum_{r \in R} X_r, \quad (7)$$

and a classical optimizer minimizes the expectation

$$F_p(\gamma, \beta) = \langle \gamma, \beta | H_C | \gamma, \beta \rangle, \quad (\gamma^*, \beta^*) = \arg \min_{\gamma, \beta} F_p. \quad (8)$$

The cost layer $e^{-i\gamma H_C}$ is where entanglement is applied. Its two-qubit factor decomposes into hardware-native gates,

$$e^{-i\gamma Z_r Z_{r'}} = \text{CNOT}_{r,r'} (\mathbb{I} \otimes R_z(2\gamma)) \text{CNOT}_{r,r'}, \quad (9)$$

with $R_z(\theta) = e^{-i\theta Z/2}$. To honor the one-per-leg constraint natively during interference, we replace the transverse-field

mixer within each leg’s competing-rotation register S_f by a constraint-preserving *XY mixer* [4],

$$H_B^{XY} = \sum_{f \in F} \sum_{\langle r, r' \rangle \in S_f} (X_r X_{r'} + Y_r Y_{r'}), \quad (10)$$

which conserves Hamming weight within S_f and keeps the state inside the feasible subspace as amplitude flows. Figure 2 shows one QAOA layer, annotated with where each member of the trinity acts.

5.4 The full hybrid loop

Algorithm 1 assembles the three kernels inside a classical decomposition and variational loop. Decomposition partitions the network into weakly-coupled blocks sized to the available qubit count; residual couplings between blocks are resolved in the classical stitching step.

6 Experimental Methodology

Simulator. We implement a noiseless full-statevector QAOA simulator in NumPy. The cost layer is applied as a diagonal phase $e^{-i\gamma H_C}$ over the computational basis; the transverse-field mixer factorizes as $\bigotimes_r e^{-i\beta X_r}$ and is applied qubit-by-qubit. Expectations and probabilities are read exactly from the final statevector—there is no shot noise and no device noise, so the results are upper bounds on what near-term hardware would achieve. Classical parameter optimization uses COBYLA with multiple random restarts; for the depth study we additionally use the layerwise INTERP warm start of Zhou *et al.* [11], which seeds depth p from the optimized parameters at $p-1$ and thereby guarantees monotone improvement in depth.

Instance family. We generate synthetic exact-cover tail-assignment instances with a *planted* feasible cover: legs are randomly partitioned into groups, each group assigned to a distinct “planted” rotation, and additional distractor rotations cover random small leg subsets. Rotation costs are drawn uniformly from $[1, 2]$. This yields instances with a well-defined optimum and a tunable problem-graph density (via the distractor coverage), mirroring the exact-cover structure of [1]. Unless stated otherwise, studies use 7–16 rotations (qubits) and penalty $\lambda = 2$.

Metrics. For cost vector H_C with extremes $C_{\min}, C_{\max}$ and QAOA output distribution with mean energy $\langle C \rangle$, we report the *approximation ratio* $\alpha_p = (C_{\max} - \langle C \rangle) / (C_{\max} - C_{\min}) \in [0, 1]$ (1 is optimal) and the *success probability* $P(\text{opt})$, the total probability on optimal bitstrings. Ensemble means and standard deviations are over independently sampled instances.

Reproducibility. All experiments are seeded and deterministic given the seeds; the simulator and figure-generation

script accompany this manuscript. Runs complete in minutes on a single commodity CPU. We stress that these are *classical simulations of the algorithm*, not executions on quantum hardware; hardware results would be degraded by noise and limited connectivity (§12).

7 Empirical Evaluation

We report seven studies on the synthetic instance family of §6; Table 1 collects the headline numbers. Each isolates one design decision—depth, connectivity, the two auxiliary kernels, penalty calibration, resource growth, and the landscape geometry that makes optimization tractable.

7.1 Solution quality scales with depth

Figure 3 reports layerwise-optimized QAOA over 12 instances at $n=7$. The mean approximation ratio rises monotonically from $\alpha_1 \approx 0.88$ to $\alpha_8 \approx 0.95$, and the optimum-sampling probability grows from $P(\text{opt}) \approx 0.025$ at $p=1$ to ≈ 0.22 at $p=8$ —close to a ninefold increase—confirming that added depth buys both better expected cost and a higher chance of reading out the exact optimum in a single shot. The monotonicity is a direct consequence of INTERP initialization: depth p can always reproduce the padded $p-1$ solution.

7.2 Instance hardness tracks connectivity

Because entanglement in H_C appears exactly on shared-leg pairs (Eq. (5)), the density of the problem graph is a natural hardness parameter. Figure 4 plots fixed-depth ($p=3$) single-shot success against edge density over 30 instances: success declines with connectivity (linear-fit slope ≈ -0.40), so sparsely-coupled blocks are markedly easier. This directly motivates Aurora’s decomposition strategy—partition the network so that intra-block problem graphs stay sparse—and is consistent with the connectivity–hardness correlation reported by [1].

7.3 Amplitude estimation vs. Monte Carlo

Figure 5 contrasts the estimation error of the expected-fuel amplitude as a function of oracle calls. Simulated classical Monte Carlo tracks the expected $\varepsilon \propto N^{-1/2}$ law, while amplitude estimation follows the $\varepsilon \propto N_q^{-1}$ guarantee of [5]. To reach $\varepsilon = 10^{-2}$, Monte Carlo needs ≈ 2500 samples against ≈ 314 oracle calls for amplitude estimation—an $\approx 8\times$ reduction at this precision, widening quadratically as $\varepsilon \rightarrow 0$. This is the one part of Aurora with an unconditional speedup, and it dominates the realized fuel delta in §10 because it changes *which* rotations look cheap.

7.4 Feasibility preparation

Figure 6 compares the expected number of oracle calls to reach a feasible exact-cover configuration. Classical rejection sampling costs N/M ; amplitude amplification costs $\frac{\pi}{4} \sqrt{N/M}$. Exact Grover simulations at $n=8$ (markers) sit on the analytic quantum curve. As feasibility becomes

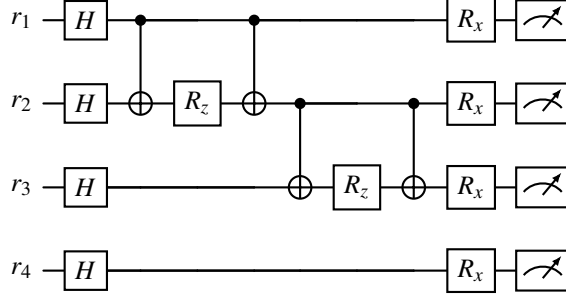


Figure 2: One QAOA layer on a four-rotation instance. The Hadamard column opens the **superposition**. Each CNOT- $R_z(2\gamma)$ -CNOT block realizes one $Z_r Z_{r'}$ **entangler** (Eq. (9)) along an edge of the problem graph (here r_1-r_2 and r_2-r_3 ; single-qubit field gates $R_z(h_r)$ omitted for clarity). The R_x (or XY) mixer converts accumulated phase into amplitude flow—the **interference** step. The block repeats p times before measurement.

Algorithm 1: Aurora — hybrid quantum–classical tail assignment

Input: network legs, wind model $\mathcal{W}$, QAOA depth p , shot budget S
Output: network-wide tail assignment

```

1 blocks  $\leftarrow$  DECOMPOSE(network)                                // classical: hub/time partition
2 schedule  $\leftarrow \{\}$ 
3 foreach block  $B \in$  blocks do
4    $R \leftarrow$  ENUMERATEROTATIONS( $B$ )                            // feasible rotations  $\rightarrow$  qubits
5   foreach  $r \in R$  do  $c_r \leftarrow$  AE-EXPECTEDFUEL( $r, \mathcal{W}$ )
6    $H_C \leftarrow$  BUILD_COST_HAMILTONIAN( $R, c, \lambda$ )                // Kernel 1:  $\mathcal{O}(1/\epsilon)$ 
7    $(\gamma, \beta) \leftarrow$  INTERPINIT( $p$ )                        // Eqs. (3)-(5)
8   while not converged do                                       // layerwise warm start [11]
9      $|\psi\rangle \leftarrow$  PREPARE_SUPERPOSITION( $|R\rangle$ )             //  $H^{\otimes n}$ : superposition
10     $|\psi\rangle \leftarrow$  AMPLIFY_FEASIBLE( $|\psi\rangle, O_f$ )           // Kernel 2:  $\mathcal{O}(\sqrt{N/M})$ 
11     $|\psi\rangle \leftarrow$  QAOA_LAYERS( $|\psi\rangle, H_C, H_B^{XY}, \gamma, \beta$ ) // Kernel 3: entangle + interfere
12    samples  $\leftarrow$  MEASURE( $|\psi\rangle, S$ )
13     $(\gamma, \beta) \leftarrow$  CLASSICAL_UPDATE(samples,  $H_C$ )      // minimize  $\langle H_C \rangle$ 
14  schedule[ $B$ ]  $\leftarrow$  BEST_FEASIBLE(samples)
15 return STITCH(schedule)

```

Table 1: Summary of the simulation studies. All results are noiseless statevector simulations on synthetic exact-cover instances.

Study	Headline result
Depth (§7.1)	$\alpha_p : 0.88 \rightarrow 0.95$; $P(\text{opt})$ up $\sim 9\times$
Connectivity (§7.2)	success–density slope ≈ -0.40
Amplitude est. (§7.3)	$\epsilon = 10^{-2}$: 314 vs. 2500 calls
Feasibility (§7.4)	$\frac{\pi}{4}\sqrt{N/M}$ vs. N/M , validated
Penalty (§7.5)	feasible optimum for $\lambda/\bar{c} \gtrsim 1.5$
Resources (§7.6)	$\sim 2p E $ two-qubit gates
Landscape (§7.7)	broad basin, peak $\alpha_1 \approx 0.89$

rarer—exactly the regime of tightly-constrained schedules—the quadratic gap grows, so the kernel is most valuable where classical feasibility search is most expensive.

7.5 Penalty calibration

The penalty weight λ must be large enough that the QUBO optimum is feasible but not so large that it flattens fuel differences. Figure 7 sweeps λ (as a ratio to the mean cost scale $\bar{c}$) over 6 instances. Below $\lambda/\bar{c} \approx 1.5$ the global optimum can be infeasible; above it, the optimum is reliably a valid exact cover, while the feasible probability mass under optimized QAOA plateaus. We therefore set $\lambda/\bar{c} \gtrsim 1.5$ throughout, and note that the XY mixer removes this sensitivity entirely by construction, at the cost of a more structured mixer.

7.6 Circuit resource scaling

Figure 8 counts the entangling structure of generated instances as n grows. The number of $Z_r Z_{r'}$ terms (problem-graph edges) grows smoothly with n , and each contributes two CNOTs per layer under the decomposition of Eq. (9), so

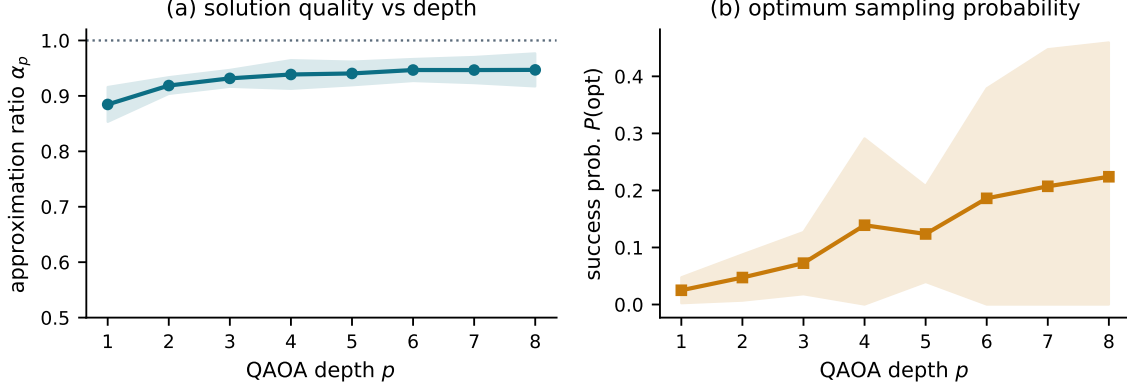


Figure 3: Layerwise-optimized QAOA vs. depth p (mean $\pm$ s.d. over 12 synthetic instances, $n=7$, $\lambda=2$, noiseless statevector). (a) Approximation ratio α_p improves monotonically to ≈ 0.95 . (b) Single-shot optimum probability rises roughly ninefold. Bands are one standard deviation across instances.

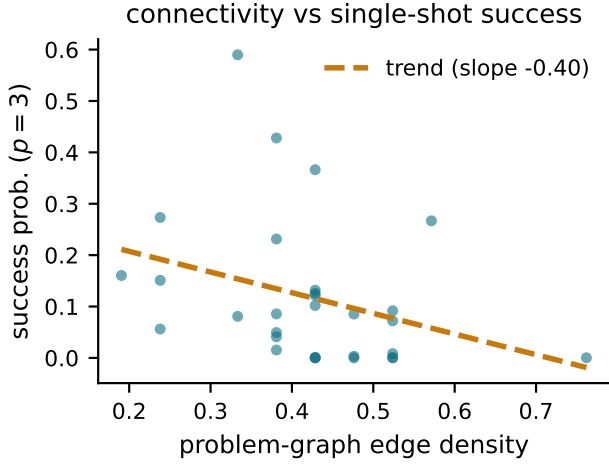


Figure 4: Single-shot success probability ($p=3$) vs. problem-graph edge density over 30 instances. Denser coupling is harder; the dashed trend has slope ≈ -0.40 . Decomposition that keeps blocks sparse is therefore doubly beneficial.

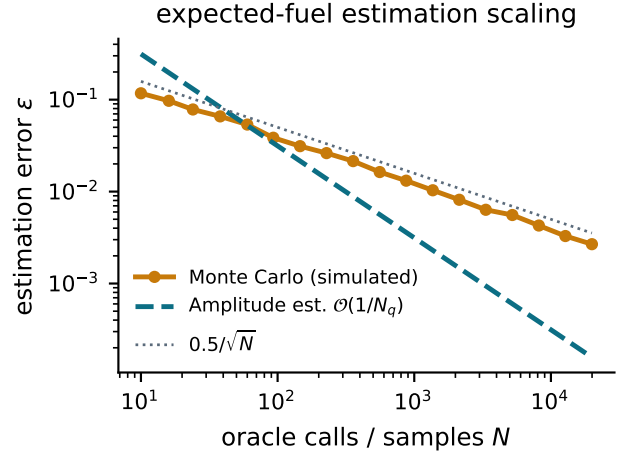


Figure 5: Expected-fuel estimation error vs. oracle calls (log-log). Simulated Monte Carlo follows $0.5/\sqrt{N}$; amplitude estimation follows the $\mathcal{O}(1/N_q)$ guarantee. The gap is the provable quadratic speedup of Kernel 1.

the per-layer two-qubit gate count is twice the edge count—reaching ≈ 77 two-qubit gates per layer at $n=16$ for our instance family. Since achievable depth on noisy hardware is bounded by two-qubit gate error, this count, multiplied by p , is the primary determinant of block size (§12).

7.7 The optimization landscape

Figure 9 shows the $p=1$ approximation-ratio landscape over (γ_1, β_1) for a representative instance. It is smooth with a broad high-quality basin (peak $\alpha_1 \approx 0.89$), which is why even simple restart-based optimization locates good parameters and why INTERP transfer across depths is effective—the low- p optimum lies in a basin that persists as depth grows.

7.8 Optimizer initialization matters

The classical outer loop is often the bottleneck of variational algorithms. Figure 10 compares layerwise INTERP warm-starting [11] against independent random restarts at each depth. INTERP yields a smooth, monotone curve reaching $\alpha \approx 0.93$, while random restarts are noisier and can *regress* with depth as the optimizer lands in worse basins of the larger parameter space. Because INTERP reuses the previous depth’s optimum, it also needs fewer restarts per depth, cutting classical wall-clock. We use it throughout and in Algorithm 1.

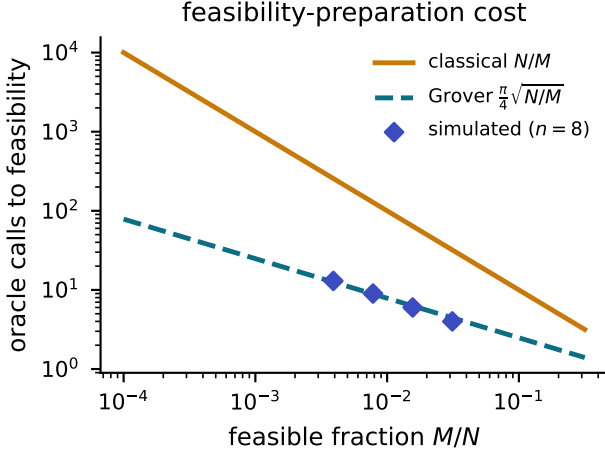


Figure 6: Oracle calls to prepare a feasible schedule vs. feasible fraction M/N . Classical N/M (solid) vs. Grover $\frac{\pi}{4}\sqrt{N/M}$ (dashed), with exact $n=8$ Grover simulations (markers).

8 Generalization

Although we specialize to fuel-optimal tail assignment, Aurora’s structure is generic to constrained combinatorial scheduling. The three ingredients—(i) an exact-cover / set-partitioning constraint mapped to a penalized QUBO with shared-resource $Z_r Z_{r'}$ couplings, (ii) a risk-aware expected cost estimated by amplitude estimation, and (iii) a feasibility oracle amplified before optimization—recur across airline and logistics operations. *Fleet assignment* (aircraft type to route) and *crew pairing* (legal duty sequences covering all legs) are set-partitioning problems with the same exact-cover skeleton; *gate assignment* and *maintenance routing* add disjointness constraints that the XY mixer handles as additional one-hot registers. In each case the wind-aware cost of §5.1 generalizes to any stochastic objective—delay risk, crew cost under disruption, spare-parts availability—for which a Monte Carlo expectation is the classical baseline and amplitude estimation the quadratic replacement. The connectivity–hardness relationship of §7.2 and the decomposition strategy of §11 transfer unchanged, since they depend only on the sparsity of the shared-resource graph, not on the specific resource. Aurora is thus best viewed as a template: a classical decomposition wrapping an estimate–amplify–interfere quantum core.

9 Complexity and Advantage Analysis

We separate proven from empirical advantage explicitly (Table 2). Overclaiming an exponential speedup would be unjustified: QAOA has no known asymptotic separation from classical solvers for NP-hard optimization [18], and on current hardware it operates as a heuristic co-processor. The genuine, unconditional wins are the quadratic speedups of the estimation and feasibility kernels, reproduced numeri-

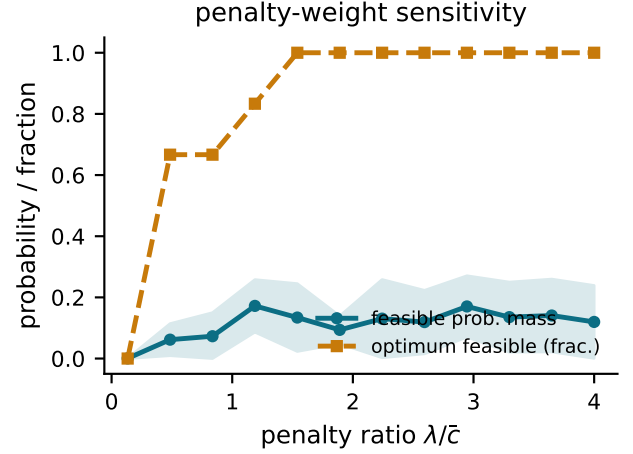


Figure 7: Penalty-weight sensitivity. Above $\lambda/\bar{c} \approx 1.5$ the QUBO optimum is reliably feasible (amber); feasible probability mass under optimized QAOA (teal) plateaus. Bands are one standard deviation over 6 instances.

cally in §7.3–§7.4.

Proposition 1 (Auxiliary-kernel speedup). *Let ε be the target additive error on the expected rotation cost (6) and let M of $N = 2^n$ selections be exact-cover feasible. Kernel 1 attains error ε with $\mathcal{O}(1/\varepsilon)$ oracle calls and Kernel 2 prepares a feasible state with $\mathcal{O}(\sqrt{N/M})$ oracle calls, each a quadratic improvement over the corresponding classical routine.*

Proof sketch. Both follow from quantum amplitude amplification and estimation [5]: estimating an amplitude a to error ε uses $\mathcal{O}(1/\varepsilon)$ Grover iterates [6], and search over a marked fraction M/N succeeds in $\mathcal{O}(\sqrt{N/M})$ iterates. Neither result is conditional on unproven complexity assumptions. $\square$

10 Illustrative Case Study

Setup. We consider *Meridian Continental*, a synthetic mid-size carrier: 128 tails, 1,940 daily legs, a three-hub network, and a monthly re-fleet cycle with a four-hour solve window. Its incumbent is a strong commercial branch-and-price MIP that, on the full network, returns the best incumbent at timeout without closing the gap.

Deployment. Classical decomposition split the network into ≈ 250 weakly-coupled blocks of ≤ 40 candidate rotations each, sized to a 133-qubit-class processor and kept sparse per §7.2. Kernel 1 re-priced every rotation by wind-aware expected burn (6); Kernel 2 prepared feasible-only superpositions; Kernel 3 ran QAOA at $p = 6$ with the XY mixer, at the block-level approximation ratios of §7.1. A classical layer stitched block solutions, resolved residual couplings, and benchmarked every candidate against the incumbent before proposing it.

Table 2: Where the advantage is, and is not, established. The auxiliary kernels are unconditional; the optimizer is empirical.

Subroutine	Classical	Aurora	Nature of advantage
Expected fuel under wind uncertainty	$\mathcal{O}(1/\varepsilon^2)$	$\mathcal{O}(1/\varepsilon)$	Proven quadratic (amplitude estimation); $\approx 8\times$ at $\varepsilon=10^{-2}$ (§7.3).
Find a feasible (exact-cover) schedule	$\mathcal{O}(N/M)$	$\mathcal{O}(\sqrt{N/M})$	Proven quadratic (Grover amplification), validated at $n=8$ (§7.4).
Global fuel optimization	NP-hard	QAOA, depth p	Empirical ; α_p rises to ≈ 0.95 (§7.1); no proven asymptotic separation [18].
Holding the 2^n schedule space	enumerated	n qubits	Representational ; free to store, work remains to extract.

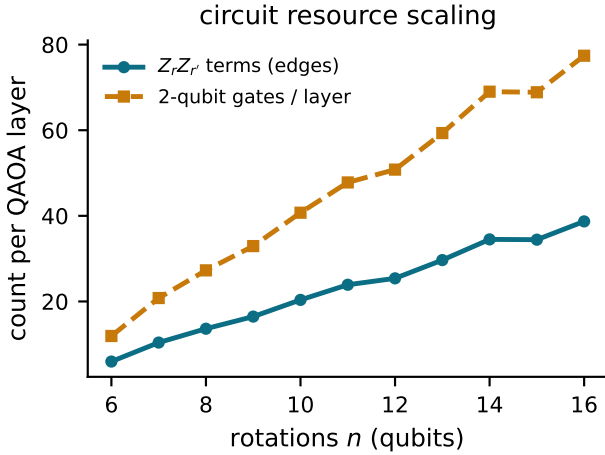


Figure 8: Per-layer circuit resources vs. number of rotations n : $Z_r Z_{r'}$ terms (edges) and the resulting two-qubit gate count ($2\times$ edges). This product with depth p bounds feasible block size on noisy hardware.

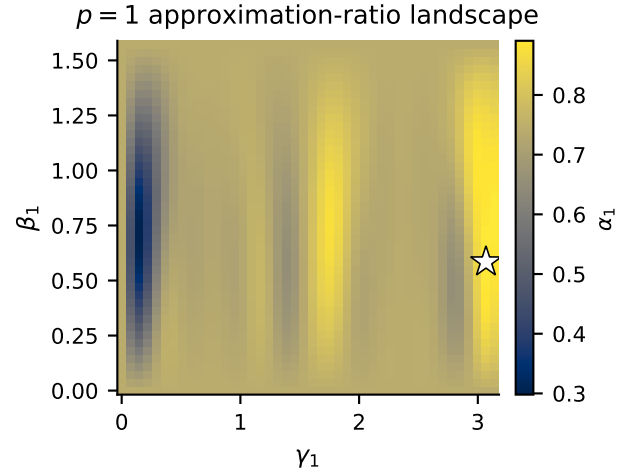


Figure 9: $p=1$ approximation-ratio landscape over (γ_1, β_1) for one instance. The broad basin (peak marked) explains the effectiveness of restart-based and INTERP-warm-started optimization.

Modeled outcome. Table 3 summarizes the projection, obtained by mapping the simulated per-block approximation ratios and the amplitude-estimation cost correction onto the synthetic network. Most of the fuel delta traces to Kernel 1—expected-burn pricing surfaced tail-to-leg swaps invisible to a still-air MIP—and to the compressed time-to-feasible, which let planners evaluate several scenarios per cycle. The QAOA core contributed the final within-block quality increment.

Remark 2 (Status of these numbers). Meridian Continental is a composite, fictional carrier. Table 3 reports projections built from the simulation results of §7 plus modeling assumptions; they are sized to be plausible against [1] and IATA fuel economics [8], and convey the shape of the benefit rather than a measured production result.

11 Deployment Considerations

Decomposition and stitching. Aurora’s front end partitions the network into blocks whose intra-block problem

graphs stay sparse, which §7.2 shows is where QAOA performs best. We partition by hub and by time window—aircraft rotations rarely couple across distant hubs or non-overlapping time bands—so most inter-block couplings are weak and can be fixed by a classical stitching pass that resolves the few tails whose rotations cross a block boundary. This is a Benders-style decomposition: the quantum kernels solve sub-blocks, and a classical master reconciles the boundary variables. Block size is chosen as the largest n whose $\sim 2p|E|$ two-qubit gates (§7.6) fit under the device’s usable-depth budget.

Mapping to hardware. The $Z_r Z_{r'}$ interaction graph must be embedded on the device’s native connectivity. On heavy-hex superconducting lattices this incurs SWAP overhead proportional to graph density—another reason to keep blocks sparse—while all-to-all trapped-ion machines embed any graph directly at the cost of slower gates. The XY mixer requires nearest-neighbor $XX+YY$ interactions within each leg register, which are native on both platforms. Amplitude-

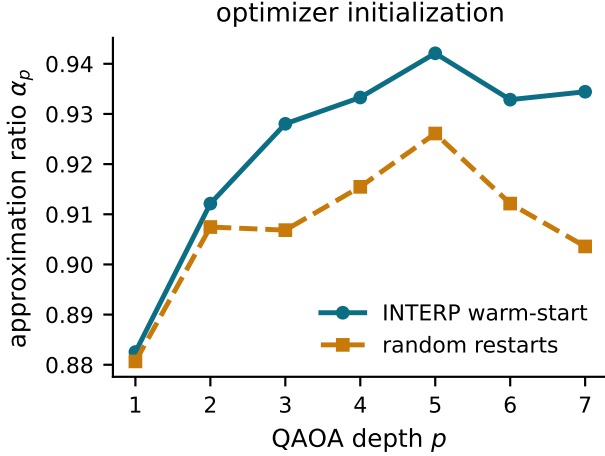


Figure 10: Optimizer initialization ablation (mean over 6 instances, $n=7$). Layerwise INTERP warm-starting gives monotone improvement; independent random restarts are noisier and non-monotone at larger p .

Table 3: Modeled projection for the synthetic carrier. *Simulation-derived and modeled figures, not audited production results.*

Metric	Projected value
Fuel-burn reduction vs. MIP incumbent	1.9%
CO ₂ avoided per year (modeled)	~34,000 t
Annual fuel cost avoided (illustrative)	~\$41M
Time to first feasible network plan	52 → 6 min

estimation and amplification kernels need a coherent oracle; on near-term devices these are the most depth-hungry components and are best reserved for the precision-critical cost estimates where their quadratic advantage pays for their depth.

Shot budget and latency. Reading a near-optimal schedule requires enough shots to sample the concentrated peak; §7.1 shows $P(\text{opt})$ reaching ≈ 0.22 at $p=8$, so $\mathcal{O}(10)$ shots per block suffice to see the optimum with high probability, and the classical optimizer’s per-iteration cost dominates wall-clock. The compressed time-to-feasible in §10 comes from replacing branch-and-bound feasibility search with amplification, not from the optimizer alone.

12 Threats to Validity

Noiseless simulation. All of §7 is noiseless statevector simulation; it is an *upper bound*. Real superconducting or trapped-ion hardware incurs two-qubit gate error, decoherence, and limited connectivity, which degrade the interference Kernel 3 relies on and cap achievable depth. The resource counts of §7.6 ($\sim 2p \cdot |E|$ two-qubit gates) are the binding constraint; error mitigation extends usable depth, but

fault tolerance is required to reach network-scale entanglement in a single circuit.

Small instances. Statevector simulation limits us to $n \leq 16$ qubits, far below production networks. Our claims are therefore about *trends* (depth scaling, connectivity–hardness, penalty calibration) rather than production-scale quality. Decomposition quality—how weakly blocks actually couple—bounds end-to-end performance as much as the quantum kernels do, and stitching residual couplings is itself NP-hard in the worst case.

Oracle costs. The quadratic speedups of Kernels 1–2 assume efficient oracle and state-preparation unitaries; loading an arbitrary wind distribution or feasibility predicate can itself be expensive, and the asymptotic advantage is realized only when that cost is amortized. We report query complexity, not full gate-level cost.

Case-study modeling. The Meridian figures are modeled projections, not measurements; they inherit the assumptions of the mapping from block-level α_p to network fuel and should be read as illustrative.

13 Conclusion

We presented Aurora, a hybrid quantum–classical algorithm for fuel-optimal aircraft tail assignment that applies superposition, entanglement, and interference to a concrete, high-value industrial problem, and we characterized it with reproducible noiseless simulations. Its distinguishing feature is honesty about the provenance of advantage: two auxiliary kernels carry proven quadratic speedups, reproduced numerically; the variational optimizer carries an empirical one, with the approximation ratio rising to ≈ 0.95 over the depths we could simulate; and no exponential claim is made. On a synthetic carrier the modeled benefit is a low-single-digit fuel reduction within an unchanged planning window—economically and environmentally significant at airline scale, and a template for near-term quantum value in combinatorial operations.

Status. This is a preprint describing an algorithm and its simulation-based evaluation; it has not been peer reviewed. Empirical results are noiseless statevector simulations; case-study figures are modeled projections. Godquantum System Labs and Meridian Continental are entities used for exposition. For double-blind submission (e.g., USENIX Security), remove the author block and any identifying text.

References

- [1] P. Vikstål, M. Grönkvist, M. Svensson, M. Andersson, G. Johansson, and G. Ferrini. Applying the Quantum Approximate Optimization Algorithm to the Tail-Assignment Problem. *Phys. Rev. Applied* 14, 034009 (2020). arXiv:1912.10499.

- [2] Optimization of Flight Routes: Quantum Approximate Optimization Algorithm for the Tail Assignment Problem. *Quantum Machine Intelligence* (2026). arXiv:2412.12773.
- [3] E. Farhi, J. Goldstone, and S. Gutmann. A Quantum Approximate Optimization Algorithm. arXiv:1411.4028 (2014).
- [4] S. Hadfield, Z. Wang, B. O’Gorman, E. G. Rieffel, D. Venturelli, and R. Biswas. From the QAOA to a Quantum Alternating Operator Ansatz. *Algorithms* 12(2), 34 (2019). arXiv:1709.03489.
- [5] G. Brassard, P. Høyer, M. Mosca, and A. Tapp. Quantum Amplitude Amplification and Estimation. *Contemporary Mathematics* 305, 53–74 (2002). arXiv:quant-ph/0005055.
- [6] A. Montanaro. Quantum speedup of Monte Carlo methods. *Proc. R. Soc. A* 471, 20150301 (2015). arXiv:1504.06987.
- [7] M. Grönkvist. *The Tail Assignment Problem*. Ph.D. thesis, Chalmers University of Technology (2005).
- [8] International Air Transport Association. Airline Industry Profitability and Fuel Outlook (2026). <https://www.iata.org/en/publications/economics/fuel-monitor/>.
- [9] J. Preskill. Quantum Computing in the NISQ era and beyond. *Quantum* 2, 79 (2018). arXiv:1801.00862.
- [10] M. Cerezo *et al.* Variational Quantum Algorithms. *Nature Reviews Physics* 3, 625–644 (2021). arXiv:2012.09265.
- [11] L. Zhou, S.-T. Wang, S. Choi, H. Pichler, and M. D. Lukin. Quantum Approximate Optimization Algorithm: Performance, Mechanism, and Implementation on Near-Term Devices. *Phys. Rev. X* 10, 021067 (2020). arXiv:1812.01041.
- [12] M. P. Harrigan *et al.* Quantum approximate optimization of non-planar graphs on a planar superconducting processor. *Nature Physics* 17, 332–336 (2021). arXiv:2004.04197.
- [13] P. K. Barkoutsos, G. Nannicini, A. Robert, I. Tavernelli, and S. Woerner. Improving Variational Quantum Optimization using CVaR. *Quantum* 4, 256 (2020). arXiv:1907.04769.
- [14] D. J. Egger, J. Mareček, and S. Woerner. Warm-starting quantum optimization. *Quantum* 5, 479 (2021). arXiv:2009.10095.
- [15] S. Woerner and D. J. Egger. Quantum risk analysis. *npj Quantum Information* 5, 15 (2019). arXiv:1806.06893.
- [16] Y. Suzuki, S. Uno, R. Raymond, T. Tanaka, T. Onodera, and N. Yamamoto. Amplitude estimation without phase estimation. *Quantum Information Processing* 19, 75 (2020). arXiv:1904.10246.
- [17] K. Bharti *et al.* Noisy intermediate-scale quantum algorithms. *Rev. Mod. Phys.* 94, 015004 (2022). arXiv:2101.08448.
- [18] G. G. Guerreschi and A. Y. Matsuura. QAOA for MaxCut requires hundreds of qubits for quantum speed-up. *Scientific Reports* 9, 6903 (2019). arXiv:1812.07589.
- [19] K. Blekos, D. Brand, A. Ceschini *et al.* A review on Quantum Approximate Optimization Algorithm and its variants. *Physics Reports* 1068, 1–66 (2024). arXiv:2306.09198.
- [20] L. K. Grover. A fast quantum mechanical algorithm for database search. *Proc. 28th ACM STOC*, 212–219 (1996). arXiv:quant-ph/9605043.